Registration No.:____________________

IMPORTANT NOTE

1) This sample paper serves solely as a visual representation of the question paper.
2) Each question in the question paper is mapped with corresponding Course Outcome(s) (CO) and Revised Bloom's Taxonomy (RBT) Level. However the question-wise CO-RBT mapping given below is indicative only and will vary in the final question paper.
3) Students should refer to the latest course syllabus and question paper pattern for exam preparation.
4) The final version of the End Term Examination question paper may differ in terms of layout, design, and content. This sample paper should be viewed for reference purpose only and not as final/official question paper.

23241CSE58039
Paper Code: A

Course Code:CSE306
Course Title:COMPUTER NETWORKS

Time Allowed: 03:00hrs. **Max Marks: 60**

Read the following instructions carefully before attempting the question paper.
1. Match the Paper Code shaded on the OMR Sheet with the Paper code mentioned on the question paper and ensure that both are the same.
2. This question paper contains 60 questions of 1 mark each. 0.25 marks will be deducted for each wrong answer.
3. Attempt all the questions in serial order.
4. Do not write or mark anything on the question paper and/or on rough sheet(s) which could be helpful to any student in copying, except your registration number on the designated space.
5. Submit the question paper and the rough sheet(s) along with the OMR sheet to the invigilator before leaving the examination hall.

Q(1) Identify the true statement out of following, if in the flag bits of IPV4, value of M (More fragments) bit is 1
(a) no more fragment
(b) last fragment
(c) can be the first one or a middle one
(d) not fragmented

CO4,L2

Q(2) To deliver a message to the correct application program running on a host, the _______ address must be consulted.
(a) IP
(b) MAC
(c) Port
(d) None of the mentioned

CO6,L4

Q(3) You are tasked with designing a network for a small office. Which routing algorithm would you use to ensure that the shortest path is always chosen, and why?
(a) Distance vector routing, because it is simple to implement.
(b) UDP, because it offers lower latency and better performance for real-time content.
(c) TCP, because it ensures that all packets arrive in order
(d) UDP, because it supports higher data integrity checks.

CO1,L1

Q(4) A user is trying to access a website by typing its URL, but the browser displays a "Server not found" error. Which of the following is the most likely cause of this issue?
(a) The user's internet connection is down
(b) The URL is incorrect.
(c) The web server is down.
(d) The domain name does not exist.

CO1,L1

Q(5) What type of cable is most commonly used for high-speed Ethernet networks?
(a) Coaxial Cable
(b) Fiber Optic Cable
(c) Twisted Pair Cable (Cat 5e/Cat 6)
(d) Wireless

CO1,L2

Q(6) Identify the correct option that involves implementing protocols or techniques to regulate the volume of data a sender can transmit before waiting for an acknowledgment.
(a) Noise control
(b) Transmission control
(c) Error control
(d) Flow control

CO3,L2

Q(7) Analyse the sequence number of the segment if all data is sent in only one segment. Suppose a TCP connection is transferring a file. The first byte is numbered 10001.
(a) 10000
(b) 12001
(c) 11001
(d) 10001

CO6,L4

Q(8) Consider a source computer(S) transmitting a file of size 106 bits to a destination computer(D)over a network of two routers (R1 and R2) and three links(L1, L2, and L3). L1connects S to R1; L2 connects R1 to R2; and L3 connects R2 to D.Let each link be of length 100 km. Assume signals travel over each link at a speed of 108 meters per second.Assume that the link bandwidth on each link is 1Mbps. Let the file be broken down into 1000 packets each of size 1000 bits. Find the total sum of transmission and propagation delays in transmitting the file from S to D?

(a) 1005ms (b) 1010ms
(c) 3000ms (d) 3003ms

CO4,L3

Q(9) Identify Which type of transmission medium uses light signals to carry data

(a) Twisted pair cable (b) Coaxial cable
(c) Fiber optic cable (d) Wireless radio waves

CO2,L2

Q(10) Which name space is preferred mostly in order to reduce ambiguity?

(a) Hierarchical namespace (b) Flat namespace
(c) Single namespace (d) None of these

CO6,L4

Q(11) What are the features included in the Distance Vector algorithm of routing?

(a) Iterative (b) Distributed
(c) Asynchronous (d) All of above

CO5,L2

Q(12) The maximum size of TCP header is ....

(a) 20 (b) 40
(c) 60 (d) 80

CO6,L4

Q(13) In random access or contention methods______________

(a) no station is superior to another station (b) none is assigned the control over another.
(c) only one station is superior (d) Both A and B

CO3,L2

Q(14) Analyze which transport layer protocol is known for providing reliable, connection-oriented communication?

(a) UDP (b) DNS
(c) FTP (d) TCP

CO6,L4

Q(15) Change in network topology requires a Link State Routing protocol to

(a) Recalculate the shortest path for all nodes.
(b) Broadcast the change to all routers to update their topology database.
(c) Reset all routing tables to default settings
(d) Immediately shut down affected routers.

CO5,L3

Q(16) Which of the following is a data link protocol?

(a) ethernet (b) point to point protocol
(c) hdlc (d) all of the mentioned

CO3,L2

Q(17) A _____ is a TCP name for a transport service access point.

(a) port (b) pipe
(c) node (d) protocol

CO6,L4

Q(18) Analyze which of the following is time sensitive application?

(a) File transfer (b) File download
(c) Internet Telephony (d) Email

CO6,L4

Q(19) Which of the following two factors, In congestion control does the performance of a network depend upon?

(a) throughput and delay
(b) amplitude and phase
(c) buffering and noise
(d) None of the above

CO5,L4

Q(20) In Link State Routing, how does a router build its routing table?

(a) By exchanging periodic updates with neighbors
(b) By flooding LSPs and calculating the shortest path
(c) By relying on a central authority
(d) By using static routes

CO5,L3

Q(21) What is the meaning of a URL?
(a) Uniform Resource Link (b) Universal Resource Locator
(c) Universal Resource Link (d) Uniform Resource Locator

CO1,L1

Q(22) Count-to-Infinity problem occurs in ....
(A) distance vector routing (B) short path first (C) link state routing (D) hierarchical routing

,L6

Q(23) Identify the correct option, when frequency change after long interval of time then that frequency will be referred as
(a) High Frequency (b) Low Frequency
(c) Zero frequency (d) Constant Frequency

CO1,L2

Q(24) Compare the functions of a hub, a switch, and a router in a network. Which device performs packet filtering?
(a) Hub (b) Switch
(c) Router (d) All of the above

CO1,L1

Q(25) In CIDR notation, what does "/24" represent in an IP address like 192.168.1.0/24
(a) The total number of subnets
(b) The number of host bits in the subnet mask
(c) The number of available host addresses in the subnet
(d) The class of the IP address

CO4,L3

Q(26) Interpret the relationship between signaltonoise ratio (SNR) and data transmission performance, and determine how an increase in SNR affects data rate.
(a) An increase in SNR improves data rate by reducing error rates and enhancing signal clarity.
(b) An increase in SNR decreases data rate due to the added complexity in signal processing.
(c) SNR has no significant impact on data rate; it only affects signal power.
(d) A higher SNR increases noise, leading to a lower data rate.

CO2,L2

Q(27) Which of the following is Interdomain routing protocol?
(a) Distance vector (b) Link State
(c) Path Vector (d) All above

CO5,L4

Q(28) Range of Class A is
(a) 1-126 (b) 128-191
(c) 1-128 (d) 1-124

CO4,L1

Q(29) Which of the following statements are TRUE?
1.TCP handles both congestion and flow control
2.UDP handles congestion but not flow control
3.Fast retransmit deals with congestion but not flow control
4.Slow start mechanism deals with both congestion and flow control
(a) S1, S3 and S4 only (b) S1, S2 and S3 only
(c) S3 and S4 only (d) S1 and S3 only

CO1,L4

Q(30) Which of the following error detection techniques can detect all single-bit errors but not all burst errors?
(a) Parity check (b) Cyclic Redundancy Check (CRC)
(c) Hamming Code (d) Checksum

CO3,L3

Q(31) Which of the following protocol is for noiseless channel?
(a) Stop and Wait ARQ (b) Go-Back-N ARQ
(c) Stop and Wait (d) Selective Repeat ARQ

CO3,L2

Q(32) Identify the term that refers to a physical layer technique
(a) FDMA (b) CDMA
(c) TDMA (d) TDM

CO1,L2

Q(33) Which one of the following algorithm is not used for congestion control?
(a) routing information protocol
(b) traffic aware routing
(c) admission control
(d) load shedding

CO4,L3

Q(34) How would you reduce the impact of collisions in a shared Ethernet network?
(a) Increase the number of parity bits
(b) Use shorter frame sizes
(c) Implement CSMA/CD
(d) Apply a higher CRC polynomial

CO3,L3

Q(35) State the correct term for the smaller units into which data is divided for transmission over a network?
(a) Frames
(b) Cells
(c) Packets
(d) Blocks

CO1,L1

Q(36) Which one of the following algorithm is not used for congestion control?
a) traffic aware routing
b) admission control
c) load shedding
d) routing information protocol

,L6

Q(37) Which of the following is an example of a digital signal?
(a) Radio waves
(b) Light waves
(c) Binary code
(d) Morse code

CO1,L2

Q(38) Deduce the sequence number of the segment if all data is sent in only one segment. Suppose a TCP connection is transferring a file. The first byte is numbered 20001.
(a) 12003
(b) 20000
(c) 20001
(d) 10002

CO6,L4

Q(39) A subset of a network that includes all the routers but contains no loops is called ________
a) spanning tree
b) spider structure
c) spider tree
d) special tree

,L6

Q(40) You are implementing a controlled access protocol in a network. Which of the following ensures that devices only transmit when permitted, and how does it manage collisions?
(a) Pure ALOHA; collisions are managed by retransmissions after random delays.
(b) CSMA/CA; devices avoid collisions by negotiating before transmitting.
(c) Token Passing; collisions are avoided by granting access to a single device at a time.
(d) CSMA/CD; collisions are detected and corrected after they occur.

CO3,L3

Q(41) Which of the following are transport layer protocols used in networking?
a) TCP and FTP
b) UDP and HTTP
c) TCP and UDP
d) HTTP and FTP

,L6

Q(42) Organize a network that requires the implementation of Ethernet technology to ensure compatibility with existing hardware. What standard should you use for Gigabit Ethernet?
(a) 10BASE-T
(b) 100BASE-TX
(c) 1000BASE-T
(d) 10GBASE-T

CO3,L3

Q(43) Which of the following is a valid subnet mask?
(a) 176.0.0.0
(b) 96.0.0.0
(c) 127.192.0.0
(d) 255.128.0.0

CO4,L3

Q(44) To detect errors in a transmitted frame, you include a 4-byte checksum value. Which error detection technique are you using?
(a) Parity Check
(b) Checksum
(c) Hamming Code
(d) CRC

CO2,L2

Q(45) In link state routing, what is the objective of the "hello" protocol?
(a) Connecting to the target
(b) Identifying neighbors and preserving link state data
(c) Sending routing updates
(d) Handling congestion

CO5,L1

Q(46) X.21 is physical level standard for
(a) X.25
(b) CSMA/CD
(c) ATM networks
(d) None of the above

CO3,L2

Q(47) In network congestion control, what does the term "back-pressure" refer to?
A. Rerouting traffic to less congested paths
B. Applying pressure on routers to increase their processing speed
C. Using feedback to slow down the sending rate
D. Forwarding packets faster to clear congestion

CO5,L2

Q(48) In an environment where signal attenuation is a concern, which factor is most crucial when selecting a transmission medium?
(a) Signal frequency
(b) Cable length
(c) Type of material used in the medium
(d) All of the above

CO2,L2

Q(49) What is the Time to Live (TTL) field in an IP header used for?
(a) To define the highest transmission unit
(b) To show the packet's priority
(c) To offer error detection
(d) To stop packets from cycling endlessly

CO4,L1

Q(50) Which error correction technique allows the receiver to correct single-bit errors and detect multiple-bit errors?
(a) Checksum
(b) Hamming code
(c) VRC (Vertical Redundancy Check)
(d) LRC (Longitudinal Redundancy Check)

CO1,L2

Q(51) How many usable IP addresses are there in a /28 subnet?
(a) 6
(b) 14
(c) 16
(d) 30

CO4,L3

Q(52) Which of the following is/are example(s) of statefull application layer protocols?
(i) HTTP
(ii) FTP
(iii) TCP
(iv) POP3
(a) (i) and (ii) only
(b) (ii) and (iii) only
(c) (ii) and (iv) only
(d) (iv) only

CO1,L1

Q(53) Data rate not depends on:
(a) bandwidth
(b) level of the signals
(c) quality of the channel
(d) none of above

CO1,L2

Q(54) Closed-Loop control mechanisms try to _________
(a) Remove after congestion occurs
(b) Remove after sometime
(c) Prevent before congestion occurs
(d) Prevent before sending packets

CO5,L4

Q(55) Which device functions in the OSI model's Data Link layer?
(a) Router
(b) Switch
(c) Modem
(d) Firewall

CO1,L1

Q(56) Determine In Dijkstra's algorithm, what is the initial state of the distances to all nodes except the source node?
(a) All distances are set to infinity except for the source node, which is zero.
(b) All distances are set to zero.
(c) All distances are set to a negative value.
(d) All distances are initially unknown.

CO5,L3

Q(57) What is the difference between baseband and broadband transmission?
(a) Baseband transmission carries multiple signals simultaneously, while broadband transmission carries a single signal.
(b) Baseband transmission carries a single signal, while broadband transmission carries multiple signals simultaneously.
(c) Baseband transmission uses analog signals, while broadband transmission uses digital signals.
(d) Baseband transmission uses digital signals, while broadband transmission uses analog signals.

CO2,L2

Q(58) Transmission data rate is decided by ____________
(a) network layer
(b) physical layer
(c) data link layer
(d) transport layer

CO1,L1

Q(59) Tell your friend Adam that which service primitive is used for block waiting for an incoming connection?
(a) Connect
(b) Listen
(c) Receive
(d) Terminate

CO1,L1

Q(60) Recognize the transmission that is exemplified by one-way communication between a radio and a listener.
(a) Half-duplex
(b) Automatic
(c) Simplex
(d) Full-Duplex

CO1,L1

--End of Question paper--